WHO WROTE IT?

Identifying LLMs from Their Responses

A matched-prompt study of CNN and recurrent neural text classifiers

Team leader	Team member	Course / date
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Abstract

This project classifies the generating LLM family from public model responses. The curated corpus contains 1,344 responses to 448 distinct questions, with exactly one answer from each of Claude, GPT, Gemini for every question. The 429 exact/near-duplicate template components remain intact across partitions. We implement a text CNN and a bidirectional LSTM, compare prompt-only, response-only and combined inputs, and repeat the required neural experiments with three initialization seeds on one fixed grouped split. Response-only CNN and BiLSTM reach 73.8% and 75.4% mean test accuracy, respectively, against a 33.3% balanced-class chance reference.

Adding the question changes mean accuracy by +2.6 percentage points for CNN and -21.2 points for BiLSTM. Prompt-only prediction acts as a controlled negative check: balanced identical questions cannot identify their three generating families. Additional experiments examine a held-out source, compact style features, formatting normalization and a 64-token input budget. The results support attribution only within the evaluated historical model versions and public benchmark distribution; they do not establish universal authorship detection.

What is included

Requirement	Evidence
CNN and RNN implementations	PyTorch CNN and packed-sequence BiLSTM; recorded configurations and checkpoints
RQ1 and RQ2	18 required neural runs: 2 models x 3 input conditions x 3 seeds
RQ3 and RQ4	Source-holdout evaluation, descriptive features, feature classifier and ablations
Reproducibility	Data manifest, fixed question IDs, metrics, predictions, histories and environment lock

All numerical results in this report are rendered from completed experiment files. No synthetic measurements or claimed student contribution split is inserted. AI assistance is disclosed with the interpretation and lessons.

1. Problem and research design

The supervised task is closed-set authorship attribution: given a question q, a response r, or both, predict one of three known LLM families. The class label is the generating family, not answer quality, factual correctness or whether the text was written by a human. Labels, generator identifiers and file metadata are excluded from the input text.

Research question	Operational comparison	Evidence required
RQ1: Who generated the response?	CNN and BiLSTM on response only	Accuracy, Macro-F1, class-level errors
RQ2: Does the question help?	Prompt / response / prompt + response, same split and seeds	Within-model differences and prompt-only control
RQ3: Does attribution transfer?	Exclude one question-source domain from training	Test on that source; qualify domain meaning
RQ4: What signals distinguish models?	Length/lexical/formatting descriptors; normalization; 64-token condition	Interpret associations and ablation sensitivity cautiously

Controls that make the comparison meaningful

Each question is paired with all three families, keeping the question distribution and class counts exactly equal. All responses to one question, and all questions in its near-duplicate template component, remain in the same split. Every neural main condition uses the same partition and initialization seed list, while its vocabulary is fit only on its corresponding training text. The response-only and combined conditions both observe up to 256 response tokens; combined input prepends up to 64 question tokens. The output-only condition answers RQ1; the three conditions jointly answer RQ2.

Hypotheses and decision rules

If response-only classification exceeds 1/3 accuracy, response-associated information helps on this benchmark. A combined-input gain suggests a possible benefit of context, but requires uncertainty analysis and cannot isolate context from changed sequence length, vocabulary and optimization. The prompt-only condition should be exactly 1/3 accurate, because a deterministic prediction for a repeated question can match exactly one of its three balanced labels. A source-holdout gap is evidence of distribution sensitivity, although a smaller training set and changed questions can also contribute.

A positive classification result does not identify a causal linguistic fingerprint by itself. Generation settings, answer length, refusal style and benchmark construction may also carry family-associated signals. Normalization and fixed token budgets probe selected alternatives without proving that a neural network uses a unique mechanism.

2. Dataset provenance and curation

We reuse existing model-output files from the public AlpacaEval repository rather than issue fresh API requests or fabricate responses. AlpacaEval was built as a model evaluation benchmark; this project repurposes its questions and generated outputs for attribution, without treating preference scores as authorship labels. Source files, upstream configurations, license notices, checksums and the transformation script are preserved in the repository.

Family label	Upstream generator	Documented output budget	Curated responses
GPT	gpt4_0613 / gpt-4-0613	max_tokens = 4096	448
Claude	claude-2	max_tokens_to_sample = 4096	448
Gemini	gemini-pro	max_output_tokens = 2048	448

Matched-question construction

Responses are joined by a normalized question key. Starting from 805 matched questions, fixed rules exclude 188 OpenAssistant-source groups and all 156 Koala-source groups. Koala is excluded from training, validation and testing to honor its source README request against training use. After source exclusions, 1 blank-answer group and 12 contact/address-pattern groups are removed (2 email, 9 phone_like, 1 street_address). These patterns can match fictional text and are screening criteria, not proof of real personal information. No answer quality or classifier score enters selection. The final 448 questions retain three nonempty responses each.

Case-insensitive model/provider substrings are replaced with the shared literal [MODEL] in 0 question groups and 5 answers (4 Claude, 1 GPT, 0 Gemini). The script confirms identical cleaned inputs across families and no canonical duplicate questions. Stable question hashes and original-text hashes preserve traceability. Near-duplicate templates are grouped separately for evaluation; the answers themselves remain untruncated in the curated dataset.

Retained prompt source / domain	Distinct prompts	Responses (3 per prompt)
helpful_base	127	381
selfinstruct	242	726
vicuna	79	237
TOTAL	448	1344

Rights and provenance

The upstream README explicitly distinguishes Apache-2.0 code and CC BY-NC 4.0 data. The processed dataset remains under the data license, with source attribution and retained notices for noncommercial academic use. The project's code license does not override those data terms. OpenAssistant-source exclusion is a conservative selection decision, not a claim that every possible source restriction has disappeared.

Different upstream prompt wrappers and output budgets are important confounds: Gemini's documented cap is smaller than GPT/Claude's, and the historical model versions were not run under one identical decoding setup. Learned distinctions may therefore reflect collection conditions as well as model behavior. Data retrieval time is distinct from generation time.

Primary source: https://github.com/tatsu-lab/alpaca_eval ; pinned revision cd543a149df89434d8a54582c0151c0b945c3d20; detailed immutable file URLs and SHA-256 hashes: data/source_manifest.json.

3. Splits, leakage prevention and controls

Main partition	Template clusters	Distinct questions	Response rows	Each family
Training	278	278	834	278
Validation	65	83	249	83
Test	86	87	261	87

The main split targets 65% training, 15% validation and 20% testing at near-duplicate template-component level, with integer rounding. Prompt fractions differ slightly because cluster sizes vary. It is generated once with split seed 2026, using original source stratification where feasible. Model seeds 42, 43 and 44 change initialization and minibatch order, not test membership. Verification rejects overlap of both prompt IDs and template components, missing groups, duplicate normalized questions under different IDs and unbalanced groups.

The fixed input-only near-duplicate rule requires at least eight word tokens, a length ratio of at least 0.60, and either character SequenceMatcher similarity of at least 0.85 or word-trigram Jaccard similarity of at least 0.80. Connected components produce 429 split groups from 448 questions; 3 groups contain multiple questions and the largest contains 15 similar templates. This is stronger than exact-question grouping but remains a heuristic rather than complete semantic deduplication.

What is fit on which data

Object	Permitted data
Token vocabulary and minimum frequency	Training text for that input condition only
TF-IDF vocabulary / IDF	Training text only
Feature scaling and classifier	Training rows only
Checkpoint selection / early stopping	Validation Macro-F1 only
Final accuracy, F1, errors and confidence intervals	Test rows after model selection
RQ4 descriptive plots	Main test responses; descriptive analysis only

Why prompt-only accuracy is exactly chance

For every held-out question, the input-only model receives the same text three times, once for each family. In evaluation mode, each repeated text produces the same prediction. Exactly one row can be correct, yielding accuracy = number of questions / (3 x number of questions) = 1/3. This is a structural negative control, not a failed effort to optimize the classifier. Macro-F1 need not equal 1/3 because prediction frequencies can be uneven.

Residual risks

Template-group splitting prevents exact and rule-detected near-duplicate question leakage, but does not guarantee that semantically similar questions cannot appear in separate partitions. The models may exploit content correctness, refusals, collection wrappers or formatting. Scrubbing explicit model/provider names removes direct labels but cannot remove every indirect identity phrase. The source-holdout study challenges one distribution shift; it cannot validate all future tasks or current model versions.

The test set is shared across seed repetitions. Seed means describe training variability on one held-out sample; they are not results from three independently sampled test sets.

4. CNN and recurrent model implementations

Component	CNN	BiLSTM (required RNN)
Embedding	Trainable 64-dimensional embedding; PAD row	Trainable 64-dimensional embedding; PAD row
Encoder	Parallel Conv1d kernels 3, 4, 5; 48 filters each; ReLU	One bidirectional LSTM; 48 hidden units per direction
Representation	Masked global max pooling; concatenate 3 x 48 = 144 values	Concatenate final forward and backward hidden states: 96 values
Padding handling	Fully contained windows; if shorter than kernel, one zero-completed window	pack_padded_sequence ignores padded suffixes
Classifier	Dropout 0.3; linear 144 -> 3 logits	Dropout 0.3; linear 96 -> 3 logits
Loss	Multiclass cross-entropy on logits	Multiclass cross-entropy on logits

Tokenization and vocabulary

A reproducible regular expression extracts word-like units and individual non-whitespace punctuation after lowercasing. The prompt-response separator is reserved as [SEP]. The vocabulary includes PAD, UNK and SEP, retains training tokens appearing at least twice, and is capped at 20,000 entries. Unseen test tokens map to UNK. Embeddings are learned from scratch, so no pretrained language model or external embedding corpus contributes to these classifiers.

Input construction and truncation

Prompt-only keeps at most 64 question tokens; response-only keeps at most 256 response tokens. Combined input concatenates the same capped 64-token question, one [SEP], and the same capped 256-token response, for at most 321 tokens. This preserves an equal response evidence budget when answering RQ2. Each batch is dynamically padded, with enough width for the largest CNN kernel. The combined model still has a longer sequence and a separately fitted vocabulary, so it is a pipeline comparison rather than a causal estimate of context alone.

Main test text	Budget	Mean original tokens	Rows truncated
Question component	64	38.7	14.9%
Response component	256	220.8	42.5%

The CNN design is inspired by convolutional sentence classification (Kim, 2014); the recurrent model uses LSTM gating (Hochreiter and Schmidhuber, 1997). This implementation and its training setup are project code, not a claimed reproduction of either paper's benchmark results.

5. Training, evaluation and uncertainty

Setting	Shared main neural protocol
Optimizer	AdamW; learning rate 0.002; weight decay 0.0001
Batch size / gradient clipping	32; global gradient norm clipped to 1.0
Maximum epochs / early stopping	20 epochs; patience 4 on validation Macro-F1
Checkpoint selection	Best validation Macro-F1, then lower validation loss in a tie; no test selection
Repetitions	Seeds 42, 43, 44; one fixed split seed 2026
Vocabulary / standard response budget	At most 20,000 vocabulary tokens; minimum count 2; first 256 answer tokens
Reference models	Majority/chance control, TF-IDF logistic regression, style-feature logistic regression

Metrics

Accuracy is the fraction of correctly classified response rows. Precision and recall are calculated separately for each family; their harmonic mean is F1. Macro-F1 averages those family F1 values with equal weight. Balanced class counts make accuracy intuitive, while Macro-F1 exposes one-family collapse. Row-normalized confusion matrices reveal which true family is misidentified as which other family. Metrics are reported as fractions in machine-readable files and percentages in this report.

Two different sources of uncertainty

The main tables show the mean and sample standard deviation across three model seeds, in percentage points. This describes optimization variation. Saved bootstrap intervals resample near-duplicate template components, carrying every question and its three answer rows together. They estimate held-out-sample uncertainty conditional on the fitted models. Resampling individual answer rows would ignore their shared-question/template dependence. Paired comparisons use the same resampled clusters for both conditions and average metric differences across matched model seeds. Exploratory intervals are not adjusted for multiple comparisons, and no confirmatory significance claim is made.

Recorded execution evidence

Response-only run	Selected epoch	Run seconds	Vocabulary	Parameters
CNN / seed 42	14	5.7	8424	576,579
BiLSTM / seed 42	4	28.8	8424	583,203

Execution used cpu with 2 PyTorch computation threads, Python 3.12.10, PyTorch 2.6.0+cpu and NumPy 2.2.6. Run seconds include training, selected-model evaluation and artifact work; they are descriptive hardware-dependent measurements.

Raw histories, predictions, configurations and run-specific metrics are stored under artifacts/runs/. Checkpoints and vocabularies for the saved seed-42 neural runs permit inference without retraining. Environment and dataset hashes are recorded alongside the results; requirements-lock.txt preserves installed package versions.

No performance target or guaranteed grade is inferred from the assignment. The evaluation emphasizes justified design, correct implementations, transparent measurements, analysis and lessons.

6. RQ1 and RQ2: required experiment results

Input	Model	Seeds	Accuracy: mean +/- SD	Macro-F1: mean +/- SD
Prompt only	CNN	3	33.3% +/- 0.0 pp	29.6% +/- 1.2 pp
Prompt only	BiLSTM	3	33.3% +/- 0.0 pp	32.2% +/- 0.6 pp
Response only	CNN	3	73.8% +/- 2.5 pp	73.5% +/- 2.4 pp
Response only	BiLSTM	3	75.4% +/- 1.5 pp	75.4% +/- 1.5 pp
Prompt + response	CNN	3	76.4% +/- 1.5 pp	76.0% +/- 1.7 pp
Prompt + response	BiLSTM	3	54.2% +/- 2.5 pp	54.1% +/- 2.4 pp

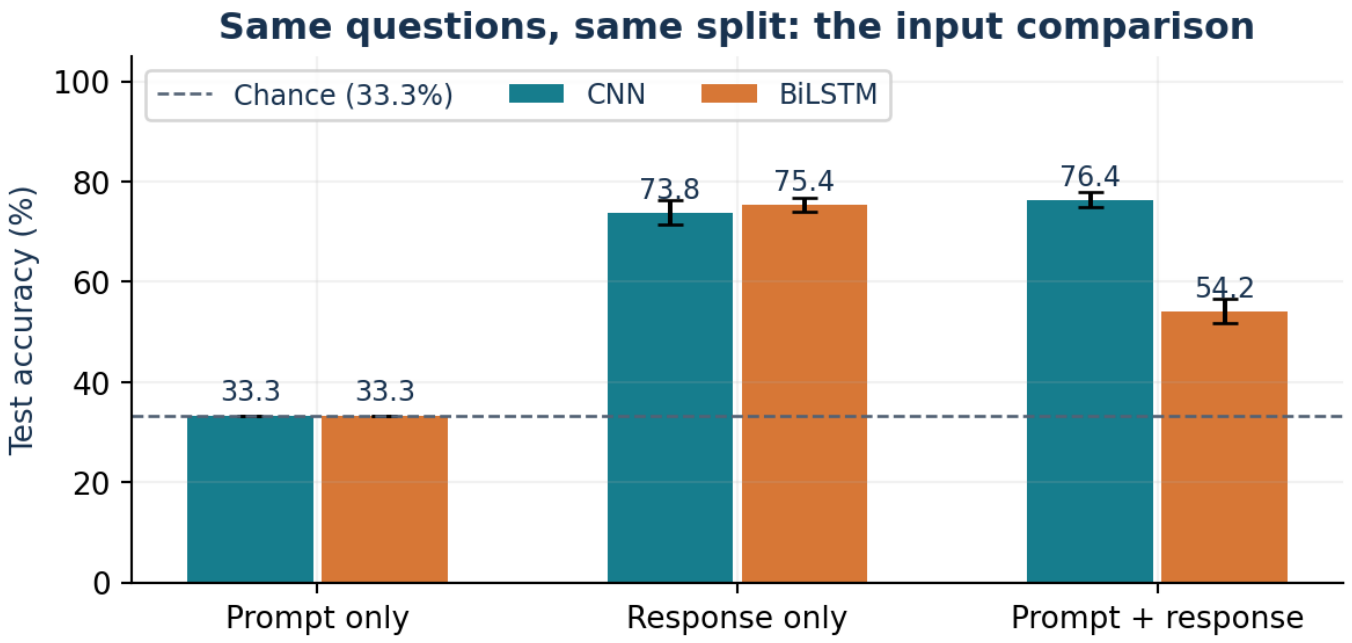


Figure 1. Means over seeds 42/43/44 on one fixed test set. Error bars are sample SD, not a confidence interval.

RQ1: response-only classification reaches 73.8% for CNN and 75.4% for BiLSTM. These are measured attribution results on the curated historical-model benchmark. RQ2: the combined condition changes CNN accuracy by +2.6 pp and BiLSTM accuracy by -21.2 pp. Prompt-only classification remains the structural chance control. Response evidence is equally capped at 256 tokens in both conditions, so a combined-score change is not caused by sacrificing response tokens to the question.

CNN context change: +2.6 pp, paired 95% interval [-0.5, +5.7] pp; the interval includes zero, so a directional difference is not established. BiLSTM context change: -21.2 pp, paired 95% interval [-26.2, -16.0] pp; the interval excludes zero in this fixed test sample. CNN pooling can retain local response patterns; the BiLSTM reverse final state summarizes toward the question prefix in combined input. Longer sequences, changed vocabulary and optimization are plausible contributors, not isolated mechanisms. Attention or response-only pooling would be useful controls.

TF-IDF and style-feature references are detailed in the appendix. They use the same split but full text rather than neural token limits, making them informative practical references rather than perfectly capacity-matched architecture controls.

7. Error structure and learning behavior

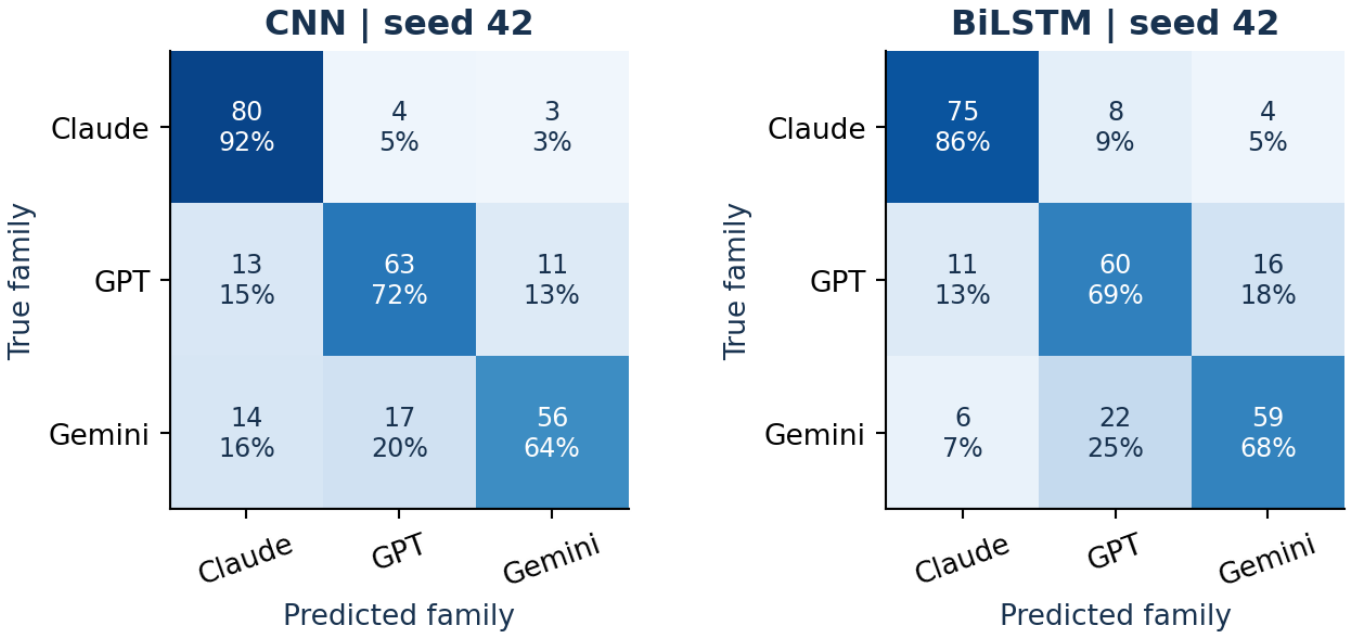


Figure 2. Response-only seed 42, selected in advance for inspection. Cells show row counts and true-class percentages; rows are actual labels, columns are predictions.

For CNN, the most frequent directional confusion is Gemini -> GPT (17 responses). The lowest class recall is Gemini at 64.4%. These are error descriptions for the prespecified representative run, not a cherry-picked best seed.

For BiLSTM, the most frequent directional confusion is Gemini -> GPT (22 responses). The lowest class recall is Gemini at 67.8%. These are error descriptions for the prespecified representative run, not a cherry-picked best seed.

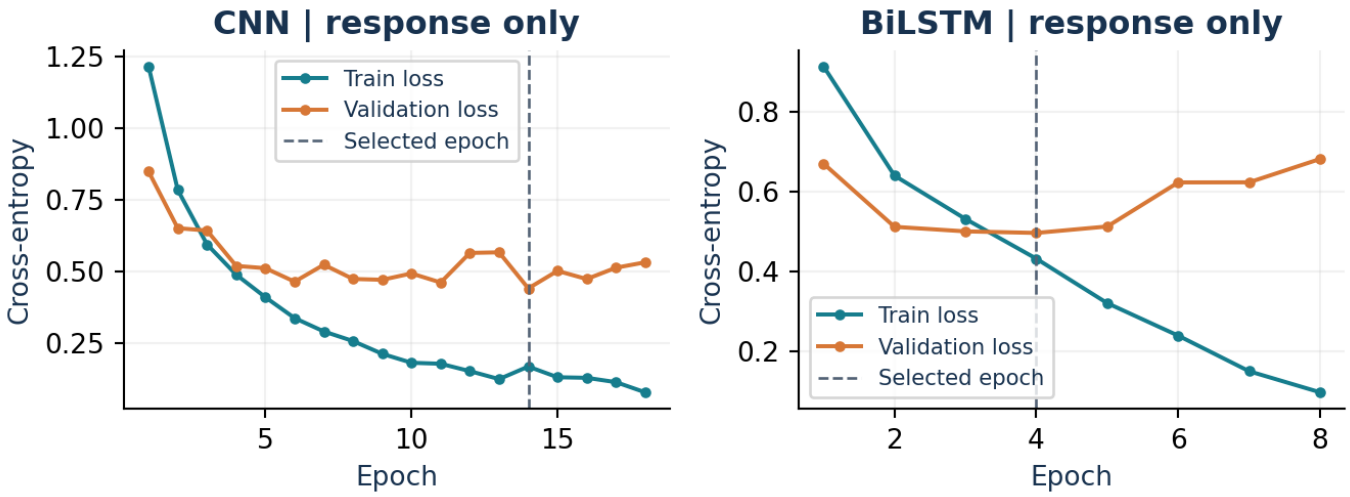


Figure 3. Actual train/validation losses for response-only seed 42. Dashed lines indicate the epoch selected by validation Macro-F1; loss itself is not the checkpoint-selection criterion.

Training loss can decrease while validation loss rises; that pattern is compatible with overfitting or changing calibration, and motivates validation-based selection. The selected checkpoint is not necessarily the minimum validation-loss epoch because the declared criterion is Macro-F1. Three seeds and a separate test partition help reveal variability but do not replace larger or external evaluations.

8. RQ3: generalization across question sources

The transfer experiment holds out the source/domain selfinstruct from training and validation. All three family responses to each held-out question form the test set. A new vocabulary and new classifier are fitted to the remaining sources. This tests a concrete source distribution shift. Upstream source names are not guaranteed to map cleanly to semantic tasks such as all coding or all writing; the result is therefore reported as source generalization rather than universal cross-task robustness.

Held-out source	Model / input	Train / val / test rows	Accuracy	Macro-F1
selfinstruct	TF-IDF / Prompt only	510 / 108 / 726	33.3%	16.7%
selfinstruct	TF-IDF / Response only	510 / 108 / 726	65.6%	65.5%
selfinstruct	TF-IDF / Prompt + response	510 / 108 / 726	64.7%	64.7%
selfinstruct	CNN / Response only	510 / 108 / 726	59.9%	59.5%
selfinstruct	BiLSTM / Response only	510 / 108 / 726	68.9%	69.0%

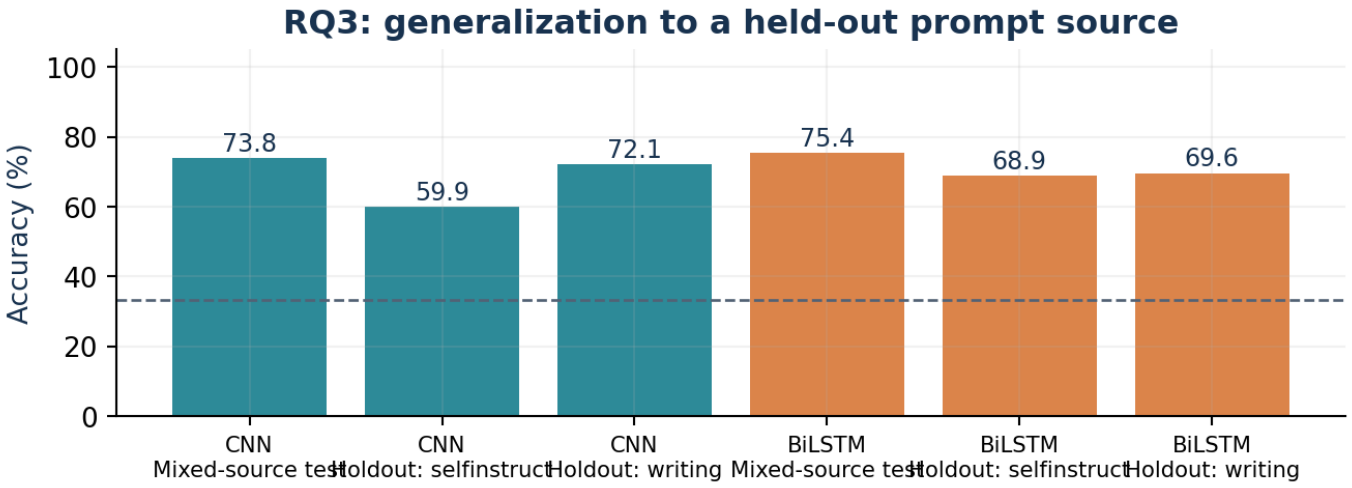


Figure 4. Main response-only accuracy averages three seeds; source-holdout runs use seed 42. Test questions and training counts differ, so this is descriptive comparison rather than a paired causal estimate.

CNN on selfinstruct differs by -13.9 percentage points from its mixed-source main mean. The held-out-source accuracy is 59.9%; the Macro-F1 is 59.5%. Differences may arise from unseen question style, topic/format composition and less training data. A smaller drop does not establish identical fingerprints across every domain.

BiLSTM on selfinstruct differs by -6.5 percentage points from its mixed-source main mean. The held-out-source accuracy is 68.9%; the Macro-F1 is 69.0%. Differences may arise from unseen question style, topic/format composition and less training data. A smaller drop does not establish identical fingerprints across every domain.

A single seed and one held-out source limit precision. An expanded study should repeat leave-one-source-out evaluation with multiple seeds, train-size-matched controls and semantically annotated task domains. The present experiment gives a reproducible, bounded answer to RQ3.

8b. RQ3: transfer to writing prompts

A second transfer study asks whether models trained on non-writing questions can identify the families on writing-like requests. Fixed ordered prompt-only keyword rules define the writing tag (for example poem, story, essay, rewrite, summarize, translate and letter). Programming, mathematics and science tags have priority in the rule order, and broad terms such as write can cause false positives. This is a reproducible approximate semantic split, not an expert-annotated clean genre partition.

Protocol	Writing holdout
Test questions / response rows	68 / 204; all three families matched per question
Training / validation questions	307 / 73; no writing test component used in fitting
Model / token-budget settings	Same main configuration; seed 42; newly fitted vocabularies and models
Selection	Validation Macro-F1 on non-writing data; no writing-test tuning

Model / input	Accuracy	Macro-F1
TF-IDF / Prompt only	33.3%	16.7%
TF-IDF / Response only	72.1%	71.6%
TF-IDF / Prompt + response	69.1%	68.3%
CNN / Response only	72.1%	72.0%
BiLSTM / Response only	69.6%	69.3%

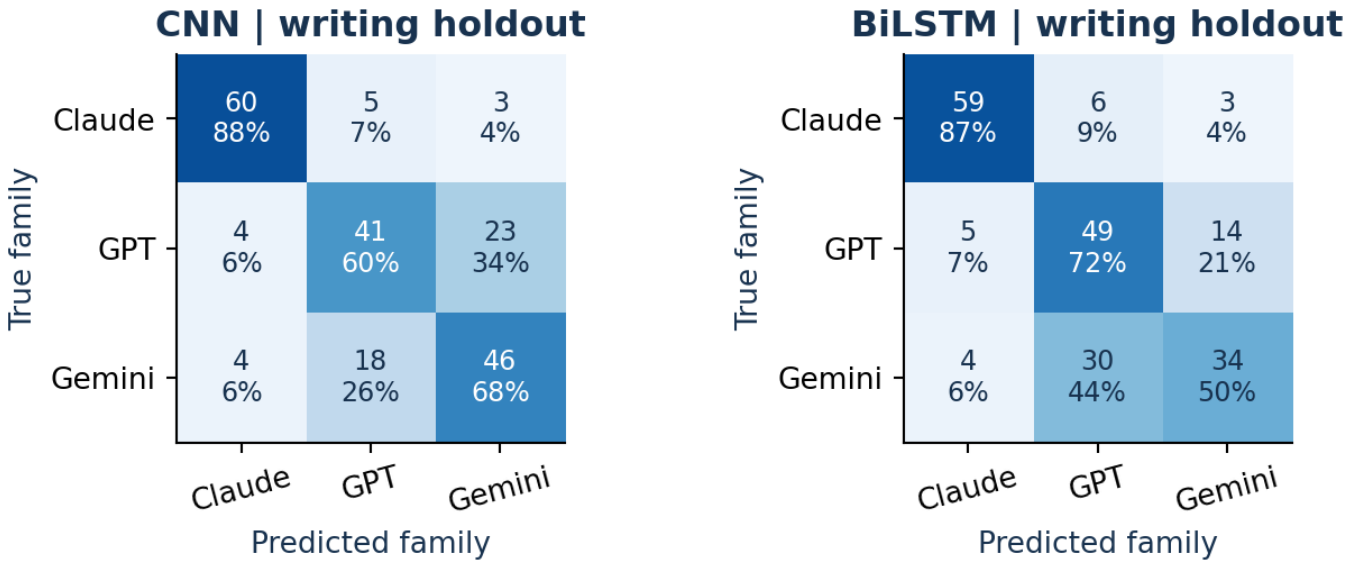


Figure 5. Seed-42 neural response-only predictions on writing-tagged questions. Counts and row percentages expose family-specific transfer errors.

CNN changes by -1.8 pp relative to the main mean; its lowest writing-test recall is GPT (60.3%). BiLSTM changes by -5.7 pp relative to the main mean; its lowest writing-test recall is Gemini (50.0%). Test populations and training sizes differ, so these gaps do not isolate task shift as the sole cause.

9. RQ4: linguistic and structural evidence

These statistics use the 261 main test responses, containing the same 87 questions for each family. Length is word count; TTR is unique lowercased words / total words. Full TTR is length-sensitive, so we also use the first 64 words (or all words for shorter answers). List/code-fence usage is the share of responses containing that format. These full-text descriptors are distinct from the neural token budgets.

Family	Mean words	Median words	Full TTR	64-word TTR	List usage	Code fences
Claude	158.7	173	0.632	0.752	62.1%	4.6%
GPT	164.7	164	0.650	0.768	46.0%	8.0%
Gemini	193.0	204	0.601	0.755	60.9%	10.3%

In this test sample, Gemini has the highest average response length (193.0 words), GPT has the highest mean 64-word-prefix TTR (0.768), and Claude most often uses lists (62.1%). Full-response TTR differences partly track length; the prefix comparison reduces but does not eliminate that confound. Code fences are rare, so their rates should not be generalized broadly.

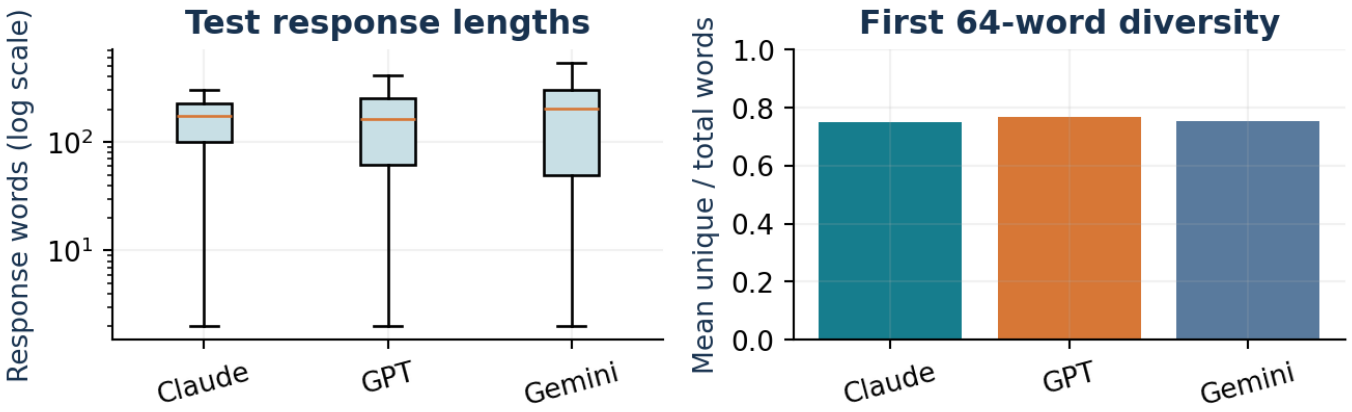


Figure 6. Test-response word-count distributions (outliers hidden in the plot only) and mean prefix lexical diversity. These describe model-associated features, not causal network attribution.

Model	Response condition (seed 42)	Accuracy	Macro-F1
CNN	Original	76.2%	75.8%
BiLSTM	Original	74.3%	74.2%
CNN	First 64 tokens	75.9%	75.4%
BiLSTM	First 64 tokens	73.9%	73.7%
CNN	Formatting normalized	66.3%	64.8%
BiLSTM	Formatting normalized	68.2%	68.0%

Each ablation retrains with seed 42 on the same split and compares against the original seed-42 run. Normalization removes Markdown/list/fence markers and collapses whitespace; it changes several properties together. The 64-token condition caps the observed prefix, while shorter-answer length remains visible. Feature-only classification and these transformations show association and sensitivity, not proof of the neural model's exact causal signals.

10. Interpretation, limitations and lessons

Paired statistical comparisons

Paired contrast	Accuracy difference	95% group-bootstrap interval
CNN: combined - response only	+2.6 pp	[-0.5, +5.7] pp
BiLSTM: combined - response only	-21.2 pp	[-26.2, -16.0] pp
Response only: CNN - BiLSTM	-1.5 pp	[-5.8, +2.6] pp
CNN: normalized - original (seed 42)	-10.0 pp	[-15.5, -4.2] pp
CNN: 64 tokens - original (seed 42)	-0.4 pp	[-4.5, +3.5] pp
BiLSTM: normalized - original (seed 42)	-6.1 pp	[-10.7, -1.5] pp
BiLSTM: 64 tokens - original (seed 42)	-0.4 pp	[-4.6, +3.7] pp

Intervals use 2,000 paired bootstrap resamples of template clusters (seed 2026). Response-only CNN minus BiLSTM: -1.5 pp, paired 95% interval [-5.8, +2.6] pp; the interval includes zero, so a directional difference is not established. CNN normalization change: -10.0 pp, paired 95% interval [-15.5, -4.2] pp; the interval excludes zero in this fixed test sample. BiLSTM normalization change: -6.1 pp, paired 95% interval [-10.7, -1.5] pp; the interval excludes zero in this fixed test sample. Intervals are exploratory and unadjusted for multiple comparisons.

What the data supports and what it cannot establish

The controlled prompt-only experiment separates question imbalance from response attribution. Response scores show that information associated with the three historical generators is recoverable in this curated benchmark. They do not separate intrinsic model style from system prompts, generation caps, preference-alignment choices, response correctness or refusal behavior. Only one version of each family is represented. The sample is modest, English-heavy and noncommercial benchmark data; similar questions and source bias may remain.

The task excludes human-written text, unregistered families, mixed-model documents, translated or edited responses and adversarial paraphrasing. A classifier must output one of its three known labels even when the real author is absent. These results should not be used as conclusive evidence about a person's writing or a current deployed model. Current-version collection, explicit unknown-class handling and independent external data would be necessary before any broader application.

10b. Lessons and experience from the project

1. Source review changed the final experiment

Reviewing the Koala README exposed a request against training use that a general benchmark license notice did not resolve. The earlier candidate subset included Koala examples. The final protocol excludes all 156 original Koala prompt groups before any split or fitting, leaving 448 matched questions. All 38 experiments were rerun, including the holdouts and ablations; earlier scores are not presented as measurements of the revised data. The verifier rejects either Koala or OASST in the submitted dataset. The lesson is to check constituent-source requests before training and to regenerate dependent results when curation changes.

2. A low control score can be the correct result

Prompt-only accuracy is exactly 33.3% across the required neural runs because each identical question occurs under all three labels. Instead of treating this as a failed classifier, the project uses it as a negative control. Grouping questions and near-duplicate templates before splitting also prevents related examples from crossing partitions. The lesson is to derive the expected control result from data construction before interpreting model accuracy; a higher prompt-only score here would require investigating imbalance or leakage.

3. Context comparisons need an equal response budget

A single total length cap would allow a prepended question to displace answer tokens, mixing the effect of context with lost response evidence. The implemented comparison retains 256 answer tokens in both conditions and adds up to 64 question tokens plus a separator for combined input. A regression test checks the shared answer budget even for a 300-token question. CNN masks padded convolution windows and BiLSTM packs sequence lengths; padding-invariance tests check that adding padded suffixes does not change predictions. The lesson is to verify input construction and batching behavior before explaining architecture differences.

4. Observed differences need uncertainty and bounded explanations

Final CNN context comparison: +2.6 pp, paired 95% interval [-0.5, +5.7] pp; the interval includes zero, so a directional difference is not established. Final BiLSTM context comparison: -21.2 pp, paired 95% interval [-26.2, -16.0] pp; the interval excludes zero in this fixed test sample. Three seeds describe initialization variability on one split. Paired bootstrap instead resamples whole test template groups. Source/writing holdouts use different questions and training sizes, and normalization changes multiple properties. The lesson is to distinguish a measured difference from its cause and to keep seed variability, test-sample uncertainty and domain shift separate.

5. Reproducibility requires checking artifacts, not only a table

The project retains original-source hashes, curated-data hashes, split membership, training histories, selected checkpoints and row-level prediction probabilities. Verification recomputes metrics for every run and replays all test rows for the 14 saved seed-42 neural checkpoints. These checks connect the manuscript to actual inference and detect stale data or model artifacts. The lesson is to treat a source-exclusion fix as a pipeline change: data, fitting, figures, interpretations, slides and the submission archive must agree.

AI assistance and responsibility

Codex assisted with data curation, implementation, experiment execution, and manuscript preparation. All reported measurements are generated by the included code; the students are responsible for reviewing, understanding, and explaining the submission.

The report records empirical design lessons rather than inventing individual student experiences or a contribution division. Both named students should review the implementation and measured outputs, correct any remaining issue and be prepared to explain the work before submission.

11. Reference classifiers and lexical associations

Reference / input	Selected C	Accuracy	Macro-F1
TF-IDF / Prompt only	0.1	33.3%	16.7%
TF-IDF / Response only	10.0	79.7%	79.6%
TF-IDF / Prompt + response	10.0	77.0%	76.9%
Style features / Response only	10.0	59.4%	59.6%
Chance / Response only	n/a	33.3%	33.4%
Majority / Response only	n/a	33.3%	16.7%

The majority baseline always selects the same training-majority family (ties resolve by fixed label order). Uniform-random accuracy is one reproducible seed-42 realization around the 1/3 expectation, so its observed value need not equal exactly 33.3%. TF-IDF extracts unigrams and bigrams, uses min_df 2, sublinear term frequency and at most 40,000 features. Logistic regularization C is chosen from 0.1, 1 and 10 by validation Macro-F1. These references use complete texts, unlike capped neural inputs.

Training-fit lexical evidence

Family	Largest positive TF-IDF coefficients
Claude	here (+6.71); here is (+5.83); here are (+4.29); are (+3.71); are some (+3.60); some (+3.50)
GPT	remember (+2.94); sure (+2.83); sure here (+2.75); might (+2.66); re (+2.25); also (+2.14)
Gemini	such (+2.00); such as (+2.00); it is (+1.96); including (+1.55); to the (+1.35); positive (+1.15)

The displayed n-grams come from the response-only TF-IDF classifier fitted on training data, after regularization selection on validation. Positive coefficients support that label relative to alternatives in this fitted classifier. Common lexical cues may reflect refusals, answer structure or dataset composition; correlated features and coefficient scale prevent treating these numbers as intrinsic provider traits or CNN/LSTM explanations.

Measurable style-feature reference

Family	Largest positive standardized feature coefficients
Claude	log_characters (+3.06); log_lines (+0.27); newline_rate (+0.23)
GPT	log_tokens (+1.97); type_token_ratio (+0.46); colon_rate (+0.13)
Gemini	asterisk_rate (+2.71); log_tokens (+0.90); mean_word_length (+0.44)

Sixteen handcrafted statistics cover log length/lines, lexical diversity, mean word length, newline/punctuation/digit rates, list-line share and code-fence count. StandardScaler is fit on training rows; coefficients are from its logistic classifier. Full-response TTR depends on length and features are correlated. The classifier's actual test score demonstrates predictive association of this feature set without establishing a causal neural mechanism.

12. Reproducibility and references

Artifact map

Artifact	Purpose
data/processed/dataset.csv	Curated balanced triples with source/domain labels
data/source_manifest.json and data/licenses/	Source URLs, checksums, filtering and applicable notices
src/llm_attribution/	Tokenizer, group splits, CNN, BiLSTM, training and inference
artifacts/results.csv and results.json	All completed experiment metrics, seed and condition labels
artifacts/runs/<run_id>/	Predictions, confusion/per-class metrics, histories, checkpoint and vocabulary where saved
artifacts/statistical_analysis.json	Paired template-cluster uncertainty estimates
report/project_report.pdf / .md	Submission report and matching readable source
docs/DEFENSE_GUIDE_KO.md	Korean explanation and presentation preparation

Reproduction

Use Python 3.12 and install requirements-lock.txt with the CPU wheel index specified in README.md. Run scripts/prepare_data.py, scripts/run_experiments.py, scripts/run_additional_domains.py (writing holdout), scripts/analyze_results.py and scripts/build_report.py, in that order. Build slides with scripts/build_slides.py and validate with scripts/verify_submission.py. README.md gives exact commands. Fixed splits and seeds are preserved; hardware/library changes can alter floating-point behavior.

Primary references

[1] CMPSC 448. Who Wrote It? Identifying LLMs from Their Responses. Assignment, modified September 17, 2026. Supplied course PDF.

[2] Li, X., Zhang, T., Dubois, Y., Taori, R., Gulrajani, I., Guestrin, C., Liang, P., and Hashimoto, T. B. (2023). AlpacaEval: An Automatic Evaluator of Instruction-following Models. GitHub repository. https://github.com/tatsu-lab/alpaca_eval . Model-output data reuse; no AlpacaEval preference evaluation is performed. Revisions are pinned in data/source_manifest.json.

[3] Kim, Y. (2014). Convolutional Neural Networks for Sentence Classification. EMNLP, 1746-1751. https://aclanthology.org/D14-1181/ . DOI: 10.3115/v1/D14-1181.

[4] Hochreiter, S., and Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation 9(8), 1735-1780. https://doi.org/10.1162/neco.1997.9.8.1735 .

[5] PyTorch 2.6 documentation. Conv1d, LSTM and pack_padded_sequence. https://docs.pytorch.org/docs/2.6/generated/torch.nn.Conv1d.html ; https://docs.pytorch.org/docs/2.6/generated/torch.nn.LSTM.html ; https://docs.pytorch.org/docs/2.6/generated/torch.nn.utils.rnn.pack_padded_sequence.html .

[6] Creative Commons. Attribution-NonCommercial 4.0 International. https://creativecommons.org/licenses/by-nc/4.0/ . Dataset notice preserved in data/licenses/.

Submission repository: https://github.com/kjk6172/cmpsc448_Midterm1. The repository must contain the source code and this PDF. Uploading files to GitHub and submitting the repository URL through the course channel are separate actions.

Dataset-source references and reuse scope

The retained question sources are credited below, following AlpacaEval's evaluation-set citation guidance. Koala is cited only to explain its exclusion from the final dataset. These references describe upstream benchmarks; their published scores are not experimental results of this project.

[7] Wang, Y., Kordi, Y., Mishra, S., Liu, A., Smith, N. A., Khashabi, D., and Hajishirzi, H. (2023). Self-Instruct: Aligning Language Models with Self-Generated Instructions. ACL (Volume 1: Long Papers), 13484-13508. https://aclanthology.org/2023.acl-long.754/ . DOI: 10.18653/v1/2023.acl-long.754.

[8] Chiang, W.-L., et al. (2023, March 30). Vicuna: An Open-Source Chatbot Impressing GPT-4 with 90%* ChatGPT Quality. LMSYS blog. https://lmsys.org/blog/2023-03-30-vicuna/ . The percentage belongs to the original blog title and is not an attribution accuracy claim.

[9] Gudibande, A., and contributors. Koala Evaluation Set. GitHub repository. https://github.com/arnav-gudibande/koala-test-set . Excluded source: its README requests no training use. No publication year or paper DOI is asserted for this repository.

[10] Anthropic. HH-RLHF dataset card and repository (helpful-base subset). https://huggingface.co/datasets/Anthropic/hh-rlhf ; https://github.com/anthropics/hh-rlhf . This project uses AlpacaEval's published questions and generator responses, not HH-RLHF preference labels.

Koala exclusion and reuse scope

Koala's README asks that its evaluation set not be used for training. All Koala-derived prompt groups are excluded from the final dataset before splitting, so none are used in training, validation or testing. The 38 final runs were retrained on the revised subset; the earlier Koala-inclusive candidate measurements are superseded. The source request and exclusion are preserved in data/NOTICE.md and the curation audit. CC BY-NC 4.0 attribution and noncommercial-use requirements still apply to the retained AlpacaEval data.

Reference metadata, immutable output-file checksums, API documentation checks and factual corrections are documented in docs/FACT_CHECK_KO.md and artifacts/reference_source_checks.json. Web access date: October 11, 2026.